

RTutor.ai Quick-Start Guide

Interaction Plots
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0) Setup

This guide uses the Grade Interaction dataset.

1. Download the Dataset
2. Upload the dataset in RTutor.ai under **(1. Data)** → **Browse**.
3. Confirm four columns: *Year-in-School*, *Major*, *Format*, *Grade*.
4. Under **(2. Data Types)** → **Edit**, set Year-in-School, Major, and Format = Category, and Grade = Integer (or Numeric).

1) Grouped Means Table

Goal: Compute the mean Grade for every combination of Format and Major. This table is the foundation for the interaction plot.

Your prompt to RTutor.ai must:

1. Reference the Grade Interaction dataset and the outcome variable Grade.
2. Ask for the mean of Grade for each combination of Format (In-Person, Online) and Major (STEM, Non-STEM). That gives four cells.
3. Request rounding to two decimal places and a clear table layout.

Check: Do you see four mean values (one for each Format × Major combination)?

2) Interaction Plot

Goal: Create an interaction plot with Format on the x-axis, mean Grade on the y-axis, and separate lines for each Major.

Your prompt must include:

1. Ask RTutor.ai to create an interaction plot (or line graph of group means) with x-axis = Format, y-axis = mean Grade, and separate lines with markers for STEM and Non-STEM.
2. Ask for “error bars” indicating the 95% margin of error.
3. Request font size 14, a color palette of your choice, and the legend at the bottom.
4. If RTutor.ai does not recognize “interaction plot,” rephrase as: “line graph of mean Grade by Format, with separate lines for each Major.”

Check: Are there two lines (STEM and Non-STEM), each connecting two points (In-Person and Online)? Does each point have bars extending above and below the point (indicating variability and uncertainty in estimation)? Look at whether the lines are roughly parallel or whether they cross/diverge.

3) Swap the Variables

Goal: Create a second interaction plot with the roles of Format and Major swapped to see the same interaction from a different angle.

1. Ask RTutor.ai for an interaction plot with x-axis = Major, y-axis = mean Grade, and separate lines for In-Person and Online.
2. Re-use the rest of your previous prompt.

Check: Both plots show the same interaction, just from different perspectives. If the lines were parallel in one plot, they should be parallel in the other. If they crossed, they should cross in both.

4) Interpretation & Extension

1. Write 2–3 sentences interpreting the interaction plot: Do the lines cross or diverge? If they are roughly parallel, the effect of Format on Grade is similar for both majors (no interaction). If they cross or diverge, the effect of Format depends on Major (interaction is present).
2. Repeat the workflow using a different pair of categorical variables (e.g., Year-in-School $\times$ Format, or Year-in-School $\times$ Major) and write the same interpretation.